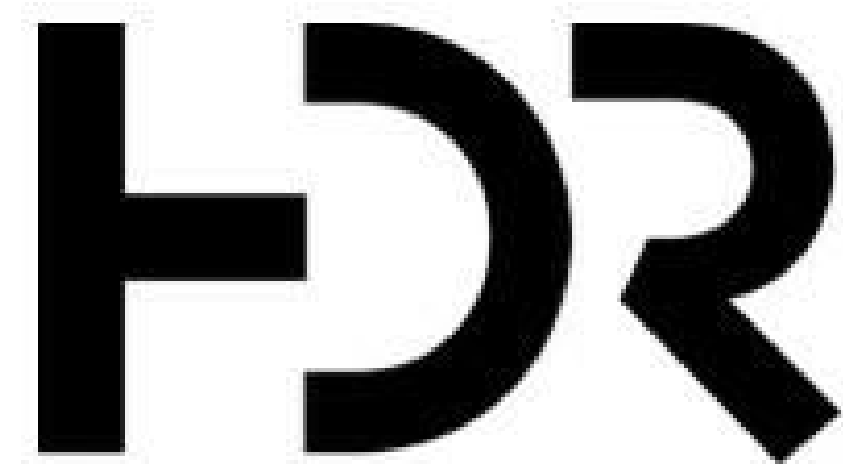




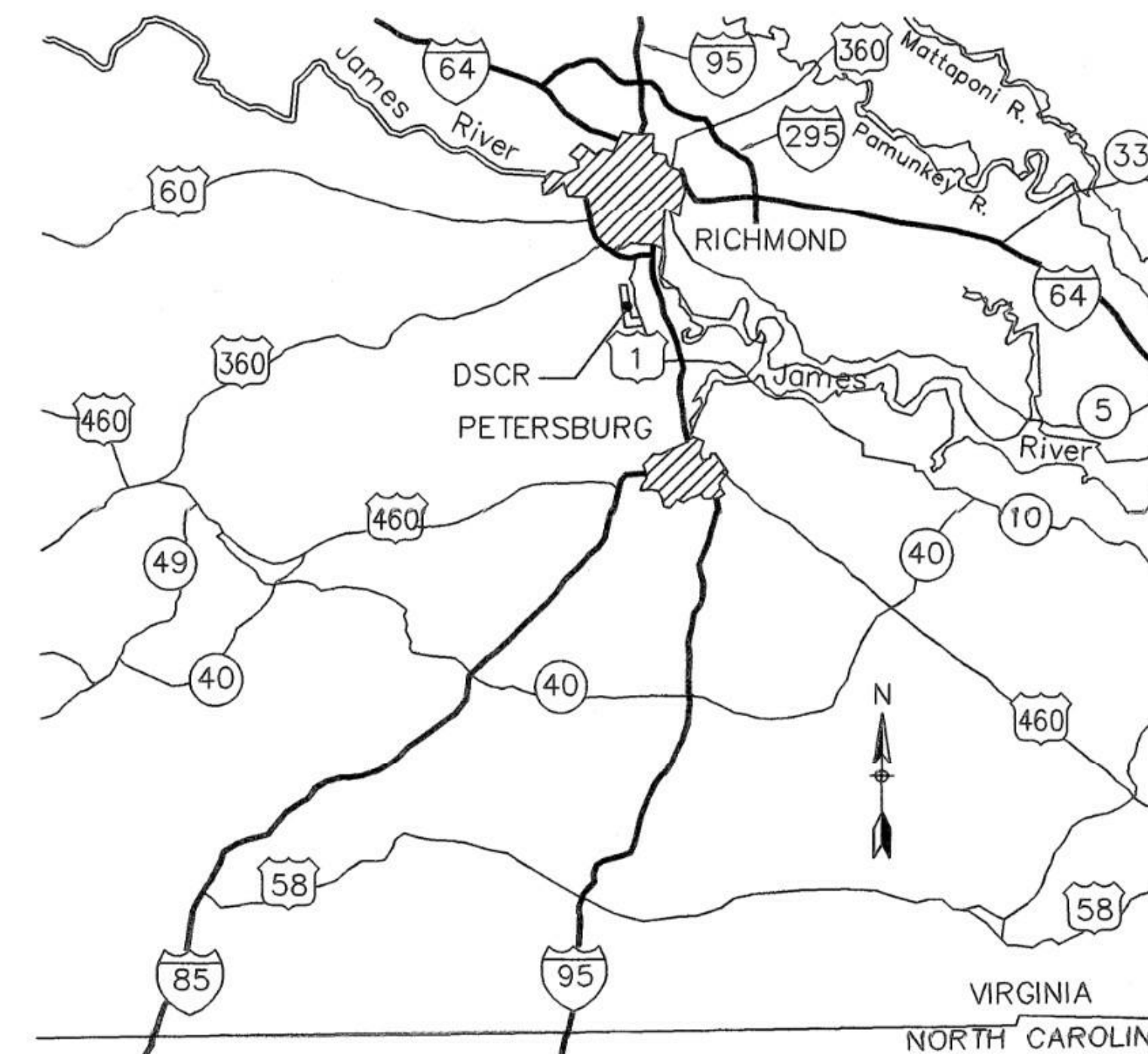
DEFENSE SUPPLY CENTER RICHMOND (DSCR)
BUILDING 46 GENERATOR TIE-IN
BLDG 46

This aerial map illustrates the 'AREA OF WORK' in the Bellwood neighborhood. The work area is delineated by a thick black dashed line, encompassing several large industrial or commercial buildings, including those numbered 3, 4, 74, 8, 9, 11, 12, 13, 14, 15, 201, 40, 44, 46, 47, 54, 59, 60, 65, 66, 80, and 98. The area is bounded by CSX Railroad tracks to the north and east, and by a residential area to the south and west. Key streets shown include 1st Street, 2nd Street, 3rd Street, 4th Street, 5th Street, 6th Street, 7th Street, 8th Street, 9th Street, 10th Street, 11th Street, 12th Street, 13th Street, 14th Street, 15th Street, 16th Street, 17th Street, 18th Street, 19th Street, 20th Street, 21st Street, 22nd Street, 23rd Street, 24th Street, 25th Street, 26th Street, 27th Street, 28th Street, 29th Street, 30th Street, 31st Street, 32nd Street, 33rd Street, 34th Street, 35th Street, 36th Street, 37th Street, 38th Street, 39th Street, 40th Street, 41st Street, 42nd Street, 43rd Street, 44th Street, 45th Street, 46th Street, 47th Street, 48th Street, 49th Street, 50th Street, 51st Street, 52nd Street, 53rd Street, 54th Street, 55th Street, 56th Street, 57th Street, 58th Street, 59th Street, 60th Street, 61st Street, 62nd Street, 63rd Street, 64th Street, 65th Street, 66th Street, 67th Street, 68th Street, 69th Street, 70th Street, 71st Street, 72nd Street, 73rd Street, 74th Street, 75th Street, 76th Street, 77th Street, 78th Street, 79th Street, 80th Street, 81st Street, 82nd Street, 83rd Street, 84th Street, 85th Street, 86th Street, 87th Street, 88th Street, 89th Street, 90th Street, 91st Street, 92nd Street, 93rd Street, 94th Street, 95th Street, 96th Street, 97th Street, 98th Street, 99th Street, 100th Street. The map also shows various landmarks such as Bensley Park, Bellwood, and the area of work. A north arrow is located in the bottom right corner.

HAUL ROUTE

SHEET LIST			65%	95%
SHEET NUMBER	SHEET NAME			
GENERAL				
G-001	COVER SHEET		•	•
STRUCTURAL				
S-001	GENERAL STRUCTURAL NOTES		•	•
SD101	FIRST FLOOR DEMOLITION PLAN			•
S-002	STRUCTURAL SPECIAL INSPECTIONS		•	•
S-003	TYPICAL CONCRETE DETAILS 1		•	•
S-100	FIRST FLOOR FRAMING PLAN		•	•
ELECTRICAL				
E-001	ELECTRICAL SYMBOLS AND GENERAL NOTES		•	•
ES101	OVERALL SITE PLAN		•	•
ES102	ENLARGED SITE PLAN		•	•
E-110	POWER PLANS		•	•
E-601	DEMOLITION ONE-LINE DIAGRAM		•	•
E-602	NEW ONE-LINE DIAGRAM		•	•

VICINITY MAP



RTA



DEFENSE SUPPLY CENTER RICHMOND (DSCR)
BUILDING 46 GENERATOR TIE-IN
10414815

COVER SHEET

SHEET ID
BLDG 46
G-001

US ARMY CORPS OF ENGINEERS
NORFOLK, VIRGINIA



DESIGNED BY:
J. GARAFOLA
DRAWN BY:
J. GARAFOLA
CHECKED BY:
J. GARAFOLA
SUBMITTED BY:

ISSUE DATE:	DECEMBER 15, 2023
SOLICITATION NO.:	FA8903-24-F-0083
CONTRACT NO.:	FA8903-21-D-0007

Plot Date:	12/15/2025 1:17:30 PM	File Path:
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P

N

M

L

K

J

H

G

F

E

D

C

B

A

GENERAL

G1. SCOPE
THE NOTES ON THIS SHEET AND THE STANDARD STRUCTURAL DETAILS ARE GENERAL AND APPLY TO THE ENTIRE PROJECT WHETHER SPECIFICALLY CALLED OUT OR NOT, EXCEPT WHERE THERE ARE SPECIFIC INDICATIONS TO THE CONTRARY ON STRUCTURAL SHEETS. IF THERE ARE QUESTIONS, THEY SHALL BE SUBMITTED TO THE STRUCTURAL ENGINEER AND ANSWERED IN WRITING PRIOR TO CONSTRUCTION.

G2. APPLICABLE SPECIFICATIONS AND CODES
A. UFC 1-200-01 (WITH CHANGE 4, DATED 17 DECEMBER 2024)
B. UFC 3-301-01 (WITH CHANGE 3, DATED 3 FEBRUARY 2025)
C. INTERNATIONAL BUILDING CODE, IBC 2024 WITH APPLICABLE EDITIONS OF THE CODE REFERENCED STANDARDS.
D. ASCE 7-22
E. ACI 318-19

G3. DESIGN CRITERIA
1. APPLIES TO ALL STRUCTURES (UNO)

A. DEAD LOAD:
 1. ACTUAL TRIBUTARY STRUCTURE WEIGHT
 2. SELF-WEIGHT OF EATON NEMA TYPE-1 TRANSFER SWITCH; 4,700 LBS

B. LIVE LOAD:

C. WIND:
 1. RISK CATEGORY: II
 2. BASIC WIND SPEED (V_{ULT}): 113 MPH
 3. ALLOWABLE STRESS WIND SPEED (V_{ASD}): 88 MPH
 4. TORNADO SPEED (V_T): NOT APPLICABLE PER ASCE 7 FIGURE 32.1-1 C

5. EXPOSURE:
 6. INTERNAL PRESSURE COEFFICIENT: ±0.18
 7. DESIGN WIND PRESSURE: 31 PSF

D. SEISMIC:
 1. RISK CATEGORY: II
 2. IMPORTANCE FACTOR: 1.0
 3. SPECTRAL RESPONSE ACCELERATION, (S_S) = 0.23g
 4. SPECTRAL RESPONSE ACCELERATION, (S_1) = 0.052g
 5. SITE CLASS: D (ASSUMED)
 6. SEISMIC DESIGN CATEGORY: B
 7. SPECTRAL RESPONSE ACCELERATION, (S_{DS}) = 0.18g
 8. SPECTRAL RESPONSE ACCELERATION, (S_{d1}) = 0.073g
 9. BASIC SEISMIC FORCE RESISTING SYSTEM: ASCE 7 CHAPTER 13 - SEISMIC DESIGN REQUIREMENTS FOR NONSTRUCTURAL COMPONENTS

10. ANALYSIS PROCEDURE: EQUIVALENT LATERAL FORCE PROCEDURE
 11. DESIGN BASE SHEAR: Cs*EQUIPMENT WEIGHT
 12. SEISMIC RESPONSE COEFFICIENT, (C_s) = 0.054
 13. COMPONENT STRENGTH FACTOR, (R_w) = 1.5

E. SNOW LOAD:
 1. GROUND SNOW LOAD (P_g) = 35 PSF
 2. FROST DEPTH = 18 IN
 3. EXPOSURE FACTOR = 0.90
 4. THERMAL FACTOR = 1.03
 5. FLAT ROOF SNOW LOAD = 30 PSF
 6. FLOOD CRITERIA: NOT APPLICABLE

G. ICE LOADS:
 1. ICE THICKNESS: 0.52 IN
 2. ICE THICKNESS FACTOR: 0.997
 3. DESIGN ICE THICKNESS: 0.518 IN
 4. DESIGN ICE WEIGTH: 2.5 PSF

H. ROOF RAIN LOAD:
 1. 15-MINUTE PRECIPITATION INTENSITY: 6.09 IN/HR
 2. 60-MINUTE PRECIPITATION INTENSITY: 3.21 IN/HR

G4. AT THE TIME OF THIS DESIGN, NO GEOTECHNICAL REPORT WAS PREPARED FOR THIS PROJECT. THE FOLLOWING ASSUMED VALVE WAS USED DURING DESIGN BASED ON THE AS-BUILT DRAWINGS OF ADJACENT STRUCTURES AND IBC TABLE 1806.2. CONTRACTOR SHALL VERIFY SOIL CONDITIONS PRIOR TO CONSTRUCTION AND PLACEMENT OF SOIL SUPPORTED CONCRETE.

ALLOWABLE SOIL BEARING PRESSURE = 1,500 PSF (GRAVITY)
 2,000 PSF (SEISMIC/WIND)
LATERAL BEARING PRESSURE = 100 PSF
COHESION = 130 PSF

G5. SAFETY
SAFETY AND STRUCTURE STABILITY DURING CONSTRUCTION ARE THE SOLE RESPONSIBILITY OF THE CONTRACTOR. STRUCTURES HAVE BEEN DESIGNED TO RESIST THE DESIGN LIVE LOADS ONLY AS A COMPLETED STRUCTURE.

G6. OPENINGS
OPENINGS FOR PIPES, DUCTS, CONDUITS, ETC. ARE NOT ALL SHOWN ON THE STRUCTURAL DRAWINGS. COORDINATE AND PROVIDE OPENINGS AS REQUIRED TO ACCOMMODATE ALL WORK SHOWN OR SPECIFIED IN THE CONTRACT DOCUMENTS AND OTHERWISE REQUIRED FOR THE FURNISHING OF A FUNCTIONALLY COMPLETE PROJECT. REINFORCE AROUND OPENINGS PER STANDARD STRUCTURAL DETAILS UNLESS OTHERWISE SHOWN.

G7. SPECIAL INSPECTIONS
SPECIAL INSPECTIONS ARE REQUIRED IN ACCORDANCE WITH CHAPTER 1 AND CHAPTER 17 OF THE IBC. PAYMENT FOR THESE INSPECTIONS IS NOT THE RESPONSIBILITY OF THE CONTRACTOR. THE CONTRACTOR SHALL PROVIDE FOR FULL ACCESS TO THE WORK BY THE SPECIAL INSPECTOR AND SHALL PROVIDE FOR THESE INSPECTIONS IN THEIR CONSTRUCTION SCHEDULE IN ACCORDANCE WITH THE SPECIFICATIONS.

G8. STANDARD DETAILS
THE STANDARD DETAILS DEPICT TYPICAL DETAILING TO BE USED ON THIS PROJECT. IF CONDITIONS ARE NOT EXPLICITLY SHOWN ON THE DRAWINGS THEY SHALL BE MADE SIMILAR TO THE STANDARD DETAILS. OBTAIN APPROVAL OF ENGINEER IN WRITING FOR SIMILAR CONDITIONS PRIOR TO CONSTRUCTION.

G9. THE CONTACTOR SHALL FIELD VERIFY ALL DIMENSIONS AND ELEVATIONS OF EXISTING CONSTRUCTION AS REQUIRED TO COORDINATE NEW CONSTRUCTION. SUBMIT REQUIRED CHANGES FOR APPROVAL.

G10. CONTRACTOR TO SUBMIT FOR REVIEW ALL EQUIPMENT SIZES, OPERATING WEIGHTS, VIBRATION FORCES, SUPPORT LOCATIONS, ALONG WITH ANY FLOOR OPENINGS, NOTCHES, AND RECESSES REQUIRED BY SUCH EQUIPMENT. CONCRETE SUPPORT PADS AND/OR FRAMING REQUIRED TO SUPPORT SAID EQUIPMENT SHALL NOT BE FABRICATED AND PLACED UNTIL THE CONCRETE SUPPORT PADS AND/OR FRAMING IS APPROVED TO SUPPORT THE EQUIPMENT.

CONSTRUCTION AT EXISTING STRUCTURES

CE1. WHERE DRILL AND EPOXY REINFORCING OR ANCHORS ARE INSTALLED, CONTRACTOR SHALL NOT DAMAGE EXISTING REINFORCING. IF REQUIRED CONTRACTOR SHALL HAVE EXISTING CONCRETE INVESTIGATED USING GROUND PENETRATING RADAR TO LOCATE EXISTING REINFORCING.

CE2. EXISTING CONCRETE SURFACES TO BE JOINED WITH NEW CONCRETE SHALL BE ROUGHENED TO A 1/4" AMPLITUDE, THOROUGHLY CLEANED BY SAND BLASTING, AND APPLY BONDING AGENT JUST PRIOR TO PLACEMENT OF NEW CONCRETE.

CE3. WHERE EXISTING REINFORCING IS TO BE PRESERVED, CONTRACTOR SHALL STRAIGHTEN AND SAND BLAST REINFORCING JUST PRIOR TO PLACEMENT OF NEW CONCRETE.

CONCRETE

C1. DESIGN STRENGTHS:
F'c = 4,500 PSI
Fy = 60,000 PSI

C2. CONCRETE COVER
UNLESS OTHERWISE NOTED, PROVIDE CONCRETE COVER FOR REINFORCING AS FOLLOWS:
CONCRETE DEPOSITED AGAINST EARTH: 3"
TIE REINFORCING AT COLS & PILASTER: 1½"
ALL OTHER: 2"
SEE DRAWINGS FOR EXCEPTIONS

C3. SEE SPECIFICATIONS FOR REINFORCING PLACEMENT REQUIREMENTS.

C4. REFER TO OTHER DISCIPLINE DRAWINGS PRIOR TO CONSTRUCTION FOR EMBEDDED ITEMS AND PENETRATIONS NOT SHOWN ON STRUCTURAL DRAWINGS. AS REQUIRED TO ACCOMMODATE ALL WORK SHOWN OR SPECIFIED IN THE CONTRACT DOCUMENTS AND OTHERWISE REQUIRED FOR THE FURNISHING OF A FUNCTIONALLY COMPLETE PROJECT. REINFORCE AROUND OPENINGS PER STANDARD STRUCTURAL DETAILS UNLESS OTHERWISE SHOWN.

C5. PROVIDE ¾" CHAMFERS AT ALL EXPOSED EDGES. NOT ALL CHAMFERS MAY BE SHOWN ON DRAWINGS.

C6. FIELD ADJUST REINFORCING AT OPENINGS AND EMBEDDED ITEMS AS INDICATED.

C7. ANCHOR BOLTS NOT SPECIFIED BY ENGINEER SHALL BE DESIGNED AND CERTIFIED BY A REGISTERED PROFESSIONAL ENGINEER, RETAINED BY THE CONTRACTOR, IN ACCORDANCE WITH APPLICABLE PROJECT AND CODE REQUIREMENTS. SUBMIT AS A SHOP DRAWING FOR REVIEW AND APPROVAL BY THE ENGINEER. COORDINATE LOCATION, SIZE AND EMBEDMENT PRIOR TO CASTING CONCRETE.

C8. ABSOLUTELY NO WELDING OF REINFORCING BARS OR TORCHING TO BEND REINFORCING BARS SHALL BE ALLOWED.

C9. CONTRACTOR SHALL SUBMIT A CONCRETE PLACEMENT PLAN IDENTIFYING JOINT TYPES, JOINT LOCATIONS AND CONCRETE PLACEMENT SEQUENCE.

C10. ALL CAST IN PLACE AND POST-INSTALLED ANCHORS INDICATED IN THE STRUCTURAL DOCUMENTS SHALL COMPLY WITH CHAPTER 17 OF ACI 318 AND CHAPTER 19 OF THE IBC. ALL EXPANSION AND ADHESIVE ANCHORS SHALL HAVE THE ICC REPORT SHOWING EQUIVALENT LOAD CAPACITY. SUBMIT AND INSTALL PER THE ICC EVALUATION REPORT.

DELEGATED DESIGN SUBMITTALS

D1. DEFINITION:

DELEGATED DESIGN SUBMITTALS ARE ITEMS FOR WHICH THE FINAL DESIGN OF A SPECIFIC ELEMENT OF THE PERMANENT CONSTRUCTION WORK EXPRESSLY DELEGATED TO A PROPERLY LICENSED AND REGISTERED DESIGN PROFESSIONAL ENGINEER RETAINED OR EMPLOYED BY THE CONTRACTOR, SUBCONTRACTOR, SUPPLIER OF ANY TIER. "DELEGATED DESIGN" APPLIES ONLY TO PERMANENT WORK THAT WILL BE PART OF THE COMPLETED PROJECT AS A FUNCTIONING WHOLE.

D2. DELEGATED DESIGN LIST:

THE FOLLOWING IS A LIST OF DELEGATED DESIGN SUBMITTALS THAT ARE EXPECTED TO CONTAIN STAMPED STRUCTURAL CALCULATIONS OF SAFETY-RELATED SYSTEM INFORMATION FOR REVIEW. THESE SUBMITTALS APPLY TO DELEGATED DESIGNED SYSTEMS PRIOR TO INSTALLATION OF THE INDICATED STRUCTURAL ELEMENT, EQUIPMENT, DISTRIBUTION SYSTEM, COMPONENT, OR ITS ANCHORAGE. THE CONTRACTOR SHALL SUBMIT THE REQUIRED ENGINEER CERTIFICATION, SUPPORTING DATA, AND STAMPED DRAWINGS FOR REVIEW AND ACCEPTANCE BY THE CONTRACTING OFFICER.

DEFERRED SUBMITTALS	
SPEC SECTIONS	ITEM
05 05 20	POST-INSTALLED CONCRETE AND MASONRY ANCHORS
05 50 13	MISCELLANEOUS METAL FABRICATIONS
05 52 00	METAL RAILINGS
VARIOUS	ANY EQUIPMENT OR COMPONENT IN WHICH A TECHNICAL SPECIFICATION REQUIRES SUBMITTAL OF EQUIPMENT OR ANCHORAGE SYSTEM CALCULATIONS

UNITED STATES OF AMERICA

ARMY CORPS OF ENGINEERS

LOGISTICS CENTER

DESIGNED BY:
C. MULDERICK

DRAWN BY:
C. MULDERICK

CHECKED BY:
K. TEEMSMA

SUBMITTED BY:

SIZE:
AISI-D

ISSUE DATE:
DECEMBER 15, 2025

SOLICITATION NO.:
FA8903-24-F-0083

CONTRACT NO.:
FA8903-21-D-0007

US ARMY CORPS OF ENGINEERS

NORFOLK, VIRGINIA

HDR

HDR ENGINEERING INC.,
613 NW LP #10, SUITE 700
SAN ANTONIO, TX 78216

Registration No. F-754

DEFENSE SUPPLY CENTER RICHMOND (DSCR)

BUILDING 46 GENERATOR TI-E-IN
10414615

GENERAL STRUCTURAL NOTES

COLORADO LICENSED
COUNTY OF MULDERICK
PROFESSIONAL ENGINEER
54037
12/15/2025

[Signature]

SHEET ID
BLDG 46
S-001

[illegible]

	US ARMY CORPS OF ENGINEERS NORFOLK, VIRGINIA		DESIGNED BY: C. MULDERICK	ISSUE DATE: DECEMBER 15, 2005
	HDPE REPAIR 613 NW LP APT. SUITE 1700 SAN ANTONIO, TX 78216		CHECKED BY: C. MULDERICK	SQUATATION NO.: 10000000000000000000
			K. TEEMSMA	CONTRACT NO.: FA8903-21-D-0007
			K. TEEMSMA	
Registration No. F-7354		SIZE: ANSI D		

DEFENSE SUPPLY CENTER RICHMOND (DSCR)
BUILDING 46 GENERATOR TIE-IN
10414815

SHEET ID
BLDG 46
S-001

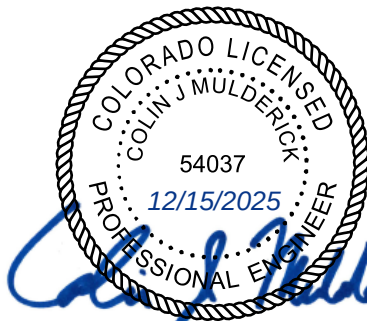
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Print Date: 12/15/2025 10:55:17 AM

Plot Date:

RTA

A circular professional engineer seal for the State of California. The outer ring contains the text "STATE OF CALIFORNIA" at the top and "PROFESSIONAL ENGINEER" at the bottom. The center of the seal features the date "12/15/2025" and a blue ink signature, "C. J. [illegible]".

12/15/2025 10:55:17 AM

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Plot Date: 12/15/2025 10:55:17 AMFile Path: C:\w\0203\104\4815_DSDG_5_BLDG_646_sdm.mxd32642.vt

RTA

STATEMENT OF SPECIAL INSPECTIONS (IBC 1705):
UNLESS NOTED OTHERWISE

SI1. SHOP FABRICATED ITEMS BY APPROVED FABRICATOR IS EXEMPT FROM SPECIAL INSPECTION. UPON COMPLETION, APPROVED FABRICATOR SHALL SUBMIT A CERTIFICATE OF COMPLIANCE STATING WORK WAS PERFORMED IN ACCORDANCE WITH APPROVED CONSTRUCTION DOCUMENTS.

SI2. CONTRACTOR SHALL SUBMIT A WRITTEN STATEMENT OF RESPONSIBILITY PRIOR TO CONSTRUCTION, INCLUDING:

A. ACKNOWLEDGEMENT OF SPECIAL INSPECTION REQUIREMENTS.

B. ACKNOWLEDGE THAT CONTROL WILL BE EXERCISED TO CONFORM WITH APPROVED CONSTRUCTION DOCUMENTS.

C. PROCEDURES FOR EXERCISING CONTROL, INCLUDING METHOD AND FREQUENCY OF REPORTING, AND DISTRIBUTION OF REPORTS.

D. IDENTIFICATION AND QUALIFICATION OF PERSON(S) EXERCISING SUCH CONTROL.

SI3. THE FOLLOWING CONSTRUCTION IS SUBJECT TO SPECIAL INSPECTION:

SCHEDULE OF SPECIAL INSPECTION SERVICES

INSPECTION ITEM REQUIRED	FREQUENCY		CODE REFERENCE	REMARKS
	CONTINUOUS	PERIODIC		
GENERAL STRUCTURAL OBSERVATIONS				
CONDUCT DAILY VISUAL OBSERVATION OF THE STRUCTURAL SYSTEMS FOR GENERAL CONFORMANCE TO THE CONSTRUCTION DOCUMENTS. PREPARE WEEKLY REPORT OF OBSERVATIONS DESCRIBING WORK PROGRESS AND NON-CONFORMING ITEMS.	-	X		
SOIL AND EARTHWORK			TABLE 1705.6	
VERIFY MATERIALS BELOW SHALLOW FOUNDATIONS ARE ADEQUATE TO ACHIEVE THE DESIGN BEARING CAPACITY.	-	X		
VERIFY EXCAVATIONS ARE EXTENDED TO PROPER DEPTH AND HAVE REACHED PROPER MATERIAL.	-	X		
PERFORM CLASSIFICATION AND TESTING OF COMPACTED FILL MATERIALS.	-	X		
DURING FILL PLACEMENT, VERIFY USE OF PROPER MATERIALS AND PROCEDURES OF THE APPROVED GEOTECHNICAL REPORT. VERIFY DENSITIES AND LIFT THICKNESSES DURING PLACEMENT AND COMPACTION OF COMPACTED FILL.	X	-		
PRIOR TO PLACEMENT OF COMPACTED FILL, INSPECT SUBGRADE AND VERIFY THAT SITE HAS BEEN PREPARED PROPERLY.	-	X		
CONCRETE AND REINFORCING STEEL			TABLE 1705.3	
INSPECT REINFORCEMENT AND VERIFY PLACEMENT.	-	X	ACI 318: CH 20, 25.2, 25.3, 26.6.1-26.6.3	
INSPECT ANCHORS CAST IN CONCRETE.	-	X	ACI 318: 26.13.3.3	
INSPECT ANCHORS POST INSTALLED IN HARDENED CONCRETE MEMBERS. A. ADHESIVE ANCHORS INSTALLED IN HORIZONTALLY OR UPWARDLY INCLINED ORIENTATIONS TO RESIST SUSTAINED TENSION LOADS B. MECHANICAL ANCHORS AND ADHESIVE ANCHORS NOT DEFINED ABOVE.	X	-	ACI 318: 26.13.3.2	
	-	X	ACI 318: 26.13.3	
VERIFY USE OF REQUIRED DESIGN MIX.	-	X	ACI 318: CH. 19, 26.4.3, 26.4.4	IBC 1904.1, 1904.2
PRIOR TO CONCRETE PLACEMENT, FABRICATE SPECIMENS FOR STRENGTH TESTS, PERFORM SLUMP AND AIR CONTENT TESTS, AND DETERMINE THE TEMPERATURE OF THE CONCRETE.	X	-	ASTM C31 ASTM C172 ACT 318: 26.5, 26.12	
INSPECT CONCRETE PLACEMENT FOR PROPER APPLICATION TECHNIQUES.	X	-	ACI 318: 26.5	
VERIFY MAINTENANCE OF SPECIFIED CURING TEMPERATURE AND TECHNIQUES.	-	X	ACI 318: 26.5.3-26.5.5	
VERIFY IN-SITU CONCRETE STRENGTH.	-	X	ACI 318: 26.11.2	
INSPECT FORMWORK SHAPE, LOCATION AND DIMENSIONS OF THE CONCRETE MEMBER BEING FORMED.	-	X	ACI 318: 26.11.1.2(b)	

US ARMY CORPS OF ENGINEERS
NORFOLK, VIRGINIA

ISSUE DATE:
DECEMBER 15, 2025
SOLICITATION NO.:
FA8903-24-F-0093
CONTRACT NO.:
FA8903-21-D-0007

DESIGNED BY:
C. MULDERICK
DRAWN BY:
C. MULDERICK
CHECKED BY:
K. TEEMSMA
SUBMITTED BY:

SIZE:
ANSI D

HDR
HDR ENGINEERING INC.
813 WILLOW LANE, SUITE 700
SAN ANTONIO, TX 78216
Texas P.E. Firm
Registration No. 17154

DEFENSE SUPPLY CENTER RICHMOND (DSCR)
BUILDING 46 GENERATOR TIE-IN
10414815

STRUCTURAL SPECIAL INSPECTIONS

SHEET ID
BLDG 46
S-002

54037
12/15/2025
PROFESSIONAL ENGINEER

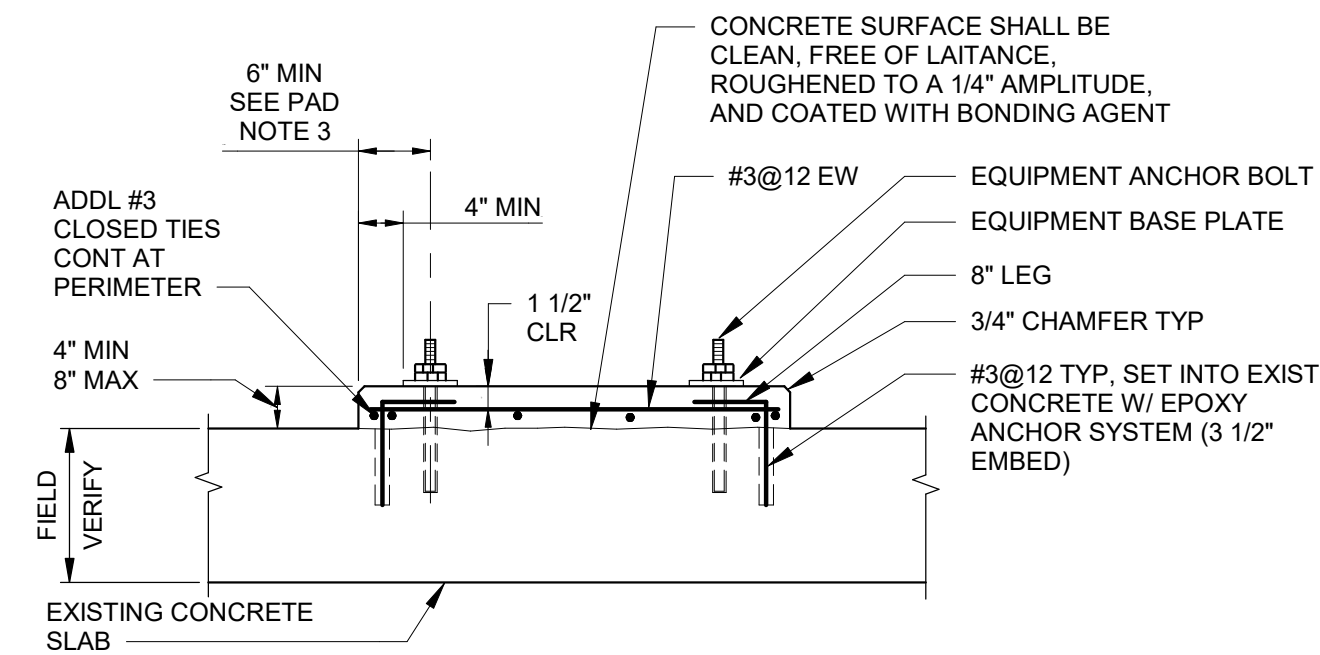
Colin J. Mulderick

LOGISTICS
DEFENSE
UNITED STATES OF AMERICA

MARK

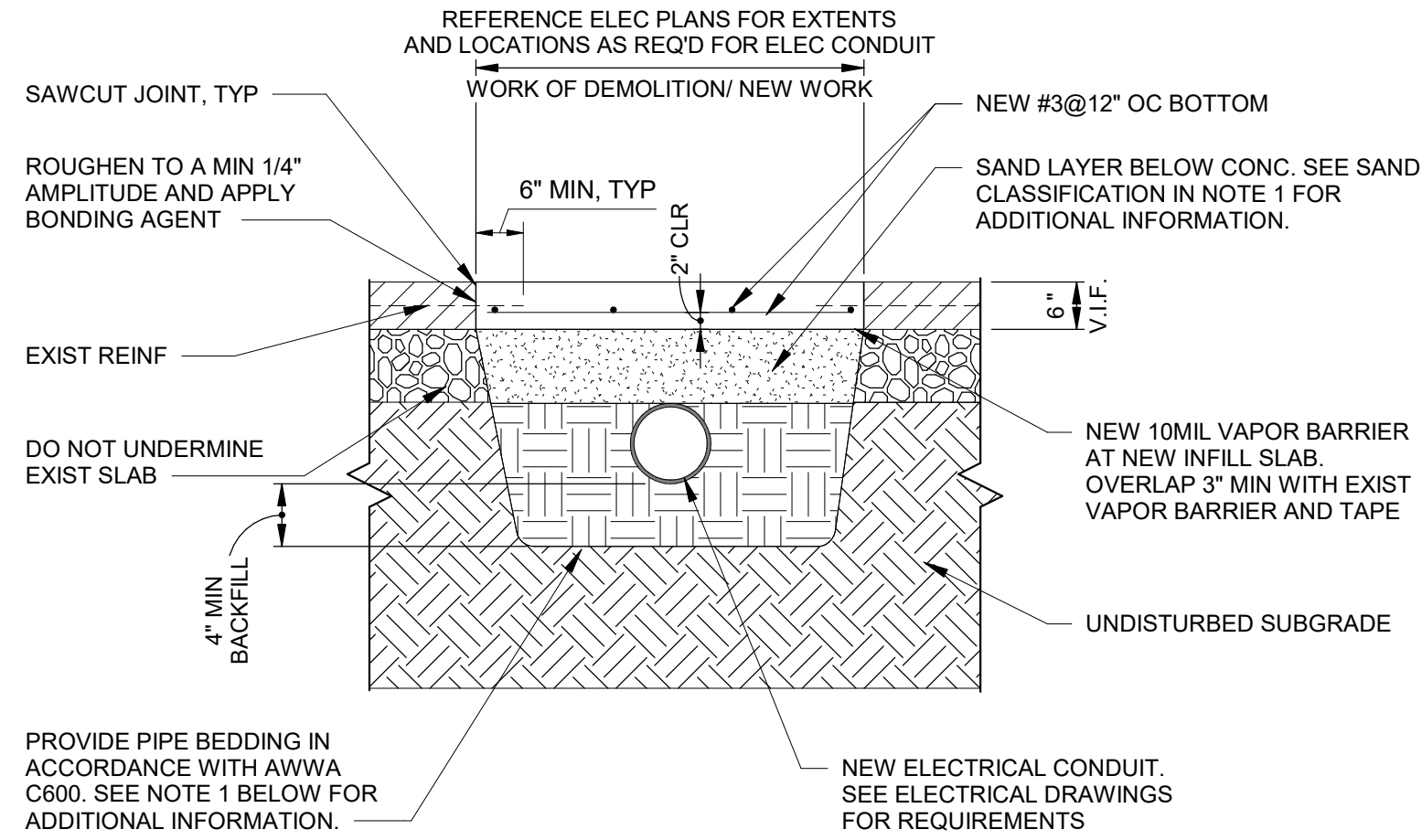
DESCRIPTION

DATE



H1) SHALLOW EQUIPMENT PAD
NTS

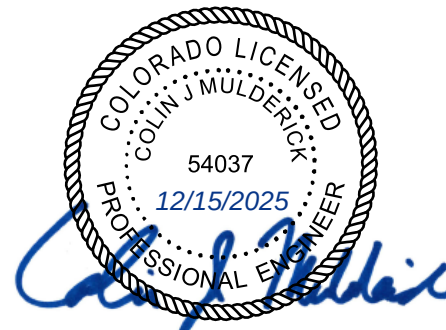
NTS



H7 SLAB INFILL FOR NEW BELOW SLAB PIPING

NTS

- NOTES:**
1. CONDUIT BEDDING MAY INCLUDED THE FOLLOWING:
 - A. CLASS I: ANGULAR, 1/4" TO 1 1/2". GRADED STONE, INCLUDING A NUMBER OF FILL MATERIALS THAT HAVE REGIONAL SIGNIFICANCE SUCH AS CORAL, SLAG, CINDERS, CRUSHED STONE, AND CRUSHED SHELLS.
 - B. CLASS II: COARSE SANDS AND GRAVELS WITH MAXIMUM PARTICLE SIZE OF 1 1/2", INCLUDING VARIOUS GRADED SANDS AND GRAVELS CONTAINING SMALL PERCENTAGES OF FINES. GENERALLY GRAVELS ARE NONCOHESIVE, EITHER WET OR DRY. SOIL TYPES GW, GP, SW, AND SP ARE INCLUDED IN THIS CLASS AS SPECIFIED IN ASTM D2487.
 - C. SAND: CLEAN, COARSE-GRAINED SAND CLASSIFIED AS SW OR SP BY ASTM D2487 FOR BEDDING BACKFILL.
 - D. GRAVEL AND CRUSHED STONE: CLEAN, COARSELY GRADED NATURAL GRAVEL, CRUSHED STONE OR A COMBINATION THEREOF IDENTIFIED AS GW OR GP IN ACCORDANCE WITH ASTM D2487 AS INDICATED.

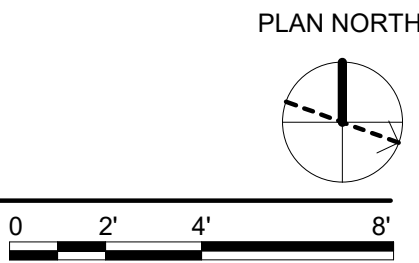


US ARMY CORPS OF ENGINEERS NORFOLK, VIRGINIA		ISSUE DATE: DECEMBER 15, 2025	
DESIGNED BY: C. MULLER/DK		SOLUTION NO.: FA8903-T-0063	
DRAWN BY: C. MULLER/DK		CHECKED BY: K. TEEMS/MSA	
DATE: 12/15/2025		FA8903-21-D-0007	
PROJECT NO.: 100		UNCLASSIFIED BY: K. TEEMS/MSA	
PROJECT NAME: SAN ANTONIO, TX 78216		SIZE: ANSI D	

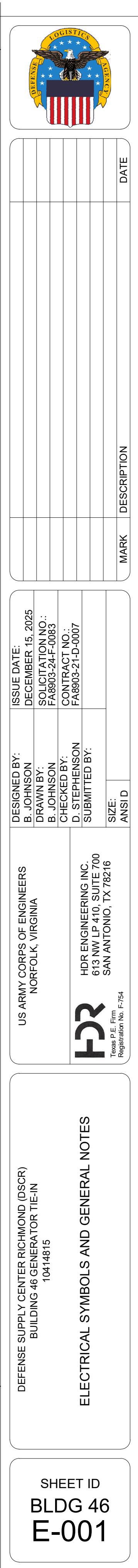
DEFENSE SUPPLY CENTER RICHMOND (DSCR)
BUILDING 46 GENERATOR TIE-IN
10414815

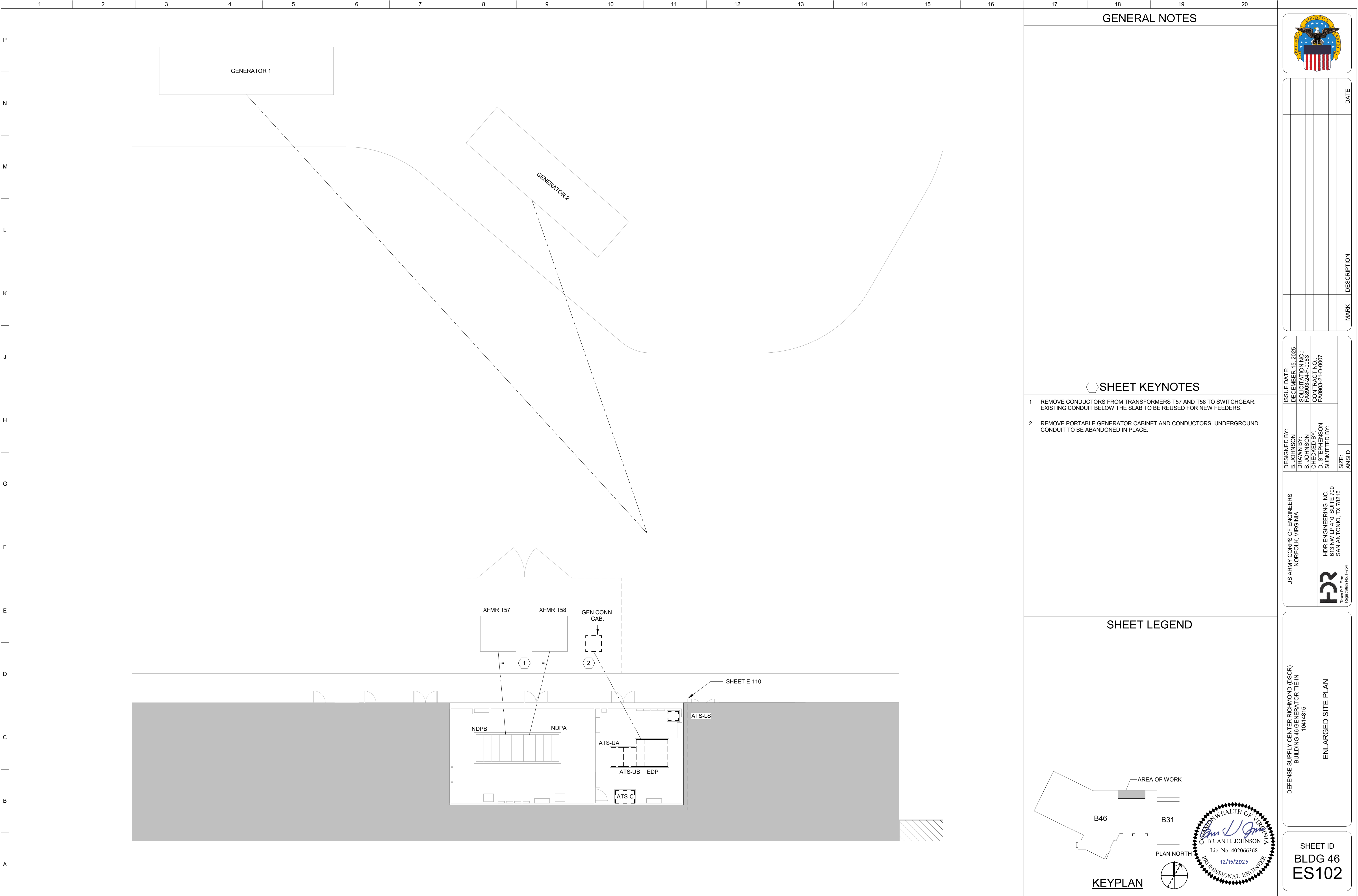
TYPICAL CONCRETE DETAILS 1

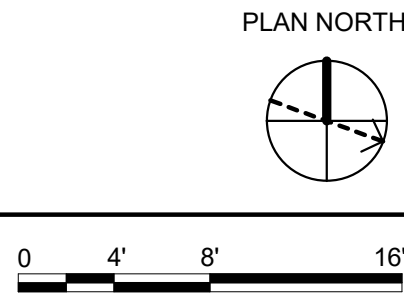
SHEET ID
BLDG 46
S-003



 HDR Headquarters Registration No. F-754	DESIGNED BY: C. MULLERBICK	ISSUE DATE: DECEMBER 15, 2025
	CHECKED BY: K. MULLERBICK	CONTRACT NO.: FA8893-21-D-0007
	DESIGNED FOR: US ARMY CORPS OF ENGINEERS NORFOLK, VIRGINIA	SUBMITTAL K. TEEMSMA
	PROJECT NO.: 613 NW PL 410, SUITE 700 SAN ANTONIO, TX 78216	SIZE: ANSI D
	DATE:	SCALE:

[illegible]





PLAN NORTH

A north arrow pointing upwards, labeled 'PLAN NORTH'. Below it is a graphic scale bar with markings for 0, 4', 8', and 16'.

- A. REFER TO STRUCTURAL DRAWINGS FOR CONCRETE WORK.
- B. PATCH/REPAIR/PAINT WALL ASSEMBLIES WHERE ELECTRICAL DEMOLITION AND NEW WORK OCCUR. MATCH EXISTING FINISH AND APPEARANCE.
- C. VERIFY THE EXACT LOCATION OF ALL UNDERGROUND CONDUITS AND UTILITIES. NOTIFY THE ENGINEER OF ANY DISCREPANCIES.
- D. ALL REMOVED EQUIPMENT SHALL BE RETURNED TO THE GOVERNMENT.
- E. EQUIPMENT SHOWN WITH DASHED LINES ARE TO BE REMOVED.
- F. EQUIPMENT SHOWN IN GREY IS EXISTING TO REMAIN.

3. ROUTE CONDUCTORS THROUGH SWITCHGEAR TO EXISTING UNDERGROUND CONDUITS.
2. ROUTE CONDUCTORS FROM TRANSFORMER IN EXISTING UNDERGROUND CONDUITS.
3. ROUTE CONDUCTORS FROM SWITCHGEAR IN EXISTING UNDERGROUND CONDUIT TO ELECTRICAL GEAR.
4. REMOVE CONDUCTORS FROM TRANSFORMERS T57 AND T58 TO SWITCHGEAR. EXISTING CONDUIT BELOW THE SLAB TO BE REUSED FOR NEW FEEDERS.
5. REMOVE CONDUCTORS FROM SWITCHGEAR TO EDP AND AUTOMATIC TRANSFER SWITCHES. CONDUITS TO BE REUSED FOR NEW FEEDERS. CONDUITS NOT REUSED ARE TO BE ABANDONED IN PLACE.
6. DISCONNECT FEEDERS FROM GENERATORS TO SWITCHBOARD EDP. FEEDERS TO BE REUSED FOR NEW AUTOMATIC TRANSFER SWITCHES.
7. PROVIDE OVERHEAD MOUNTED BUSWAY FROM ATS TO SWITCHGEAR. BUSWAY-A AND BUSWAY-B TO BE MOUNTED ONE ON TOP OF THE OTHER AND ARRANGED SO THAT THE PHASES ARE DIRECTLY IN LINE WITH THE BREAKER LUGS. PROVIDE ALL PARTS AND PIECES NECESSARY FOR A COMPLETE AND OPERATIONAL INSTALLATION. REFER TO ARCHITECTURAL PLANS FOR WALL PENETRATION.
8. PROVIDE WIREWAY TO ROUTE CONDUCTORS TO PANEL 1EPLS. EXTEND UNDERGROUND CONDUITS UP TO THE WIREWAY AND ROUTE CONDUCTORS TO 1EPLS.
9. PROVIDE OVERHEAD FEEDERS FROM EXISTING BREAKERS TO EXISTING PANELBOARDS. REFER TO ONE-LINE DIAGRAM FOR MORE INFORMATION.
10. ROUTE CONDUCTORS THROUGH SWITCHGEAR AND UP THROUGH THE SECTION OVERHEAD TO ATS-B.
11. PROVIDE ELBOW UNDERGROUND TO MAKE CONDUITS CONTINUOUS.
12. PROVIDE OVERHEAD FEEDERS FROM SWITCHGEAR TO ATS-B.
13. FIELD VERIFY GENERATOR CONDUCTOR LENGTH TO TERMINATE ON NEW EQUIPMENT. SPLICE AND EXTEND FEEDERS AS NEEDED.

AREA OF WORK

B46

B31

PLAN NORTH

KEYPLAN

[illegible]

DESIGNED BY:	B. JOHNSON	ISSUE DATE:	DECEMBER 15, 2025
DRAWN BY:	B. JOHNSON	SOLICITATION NO.:	FA8903-24-F-0083
CHECKED BY:	D. STEPHENSON	CONTRACT NO.:	FA8903-21-D-0007
SUBMITTED BY:			
SIZE:			
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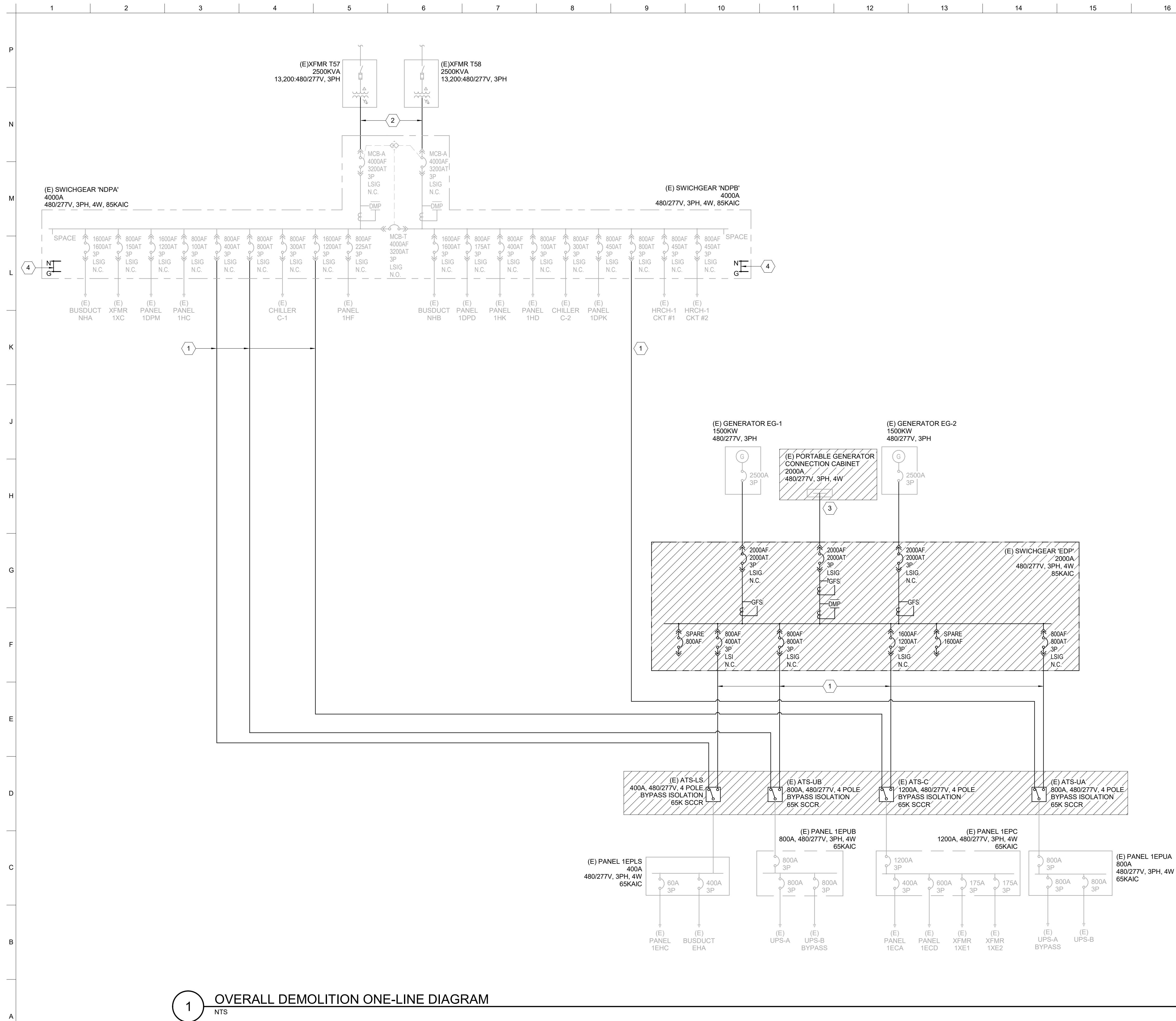
HDR ENGINEERING INC.
613 NW LP 410, SUITE 7
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Texas P.E. Firm
Registration No. F-754

US ARMY CORPS OF ENGINEERS
NORFOLK, VIRGINIA

DEFENSE SUPPLY CENTER RICHMOND (DSCR)
BUILDING 46 GENERATOR TIE-IN
10414815

SHEET ID
BLDG 46
E-110



GENERAL NOTES

- A. CONTRACTOR TO PERFORM PHASING AND WORK IN A WAY TO SHUTDOWN INDIVIDUAL FEEDER BREAKERS IN SWITCHGEAR NDPA OR NDPB TO THE GREATEST EXTENT POSSIBLE.
- B. WHEN SWITCHGEAR NDPA OR NDPB REQUIRES COMPLETE SHUTDOWN, ALL WORK MUST BE COMPLETED AFTER HOURS. FOR EXTENDED SHUTDOWNS, THE CONTRACTOR IS REQUIRED TO COORDINATE WITH THE GOVERNMENT TO PERFORM WORK ON WEEKENDS AND/OR HOLIDAYS.

[illegible]

SHEET KEYNOTES

- 1 REMOVE CONDUCTORS FROM SWITCHGEAR TO EDP AND AUTOMATIC TRANSFER SWITCHES. CONDUITS TO BE REUSED FOR NEW FEEDERS. CONDUITS NOT REUSED ARE TO BE ABANDONED IN PLACE.
- 2 REMOVE CONDUCTORS FROM TRANSFORMERS T57 AND T58 TO SWITCHGEAR. EXISTING CONDUIT BELOW THE SLAB TO BE REUSED FOR NEW FEEDERS.
- 3 REMOVE PORTABLE GENERATOR CABINET AND CONDUCTORS. UNDERGROUND CONDUIT TO BE ABANDONED IN PLACE.
- 4 REMOVE EXISTING NEUTRAL-GROUND BOND.

SUBMITTED BY:	D. STEPHENSON
CHECKED BY:	B. JOHNSON
DRAWN BY:	B. JOHNSON
DESIGNED BY:	B. JOHNSON
SIZE:	ANSI D
SUBMITTED BY:	D. STEPHENSON
CHECKED BY:	B. JOHNSON
DRAWN BY:	B. JOHNSON
DESIGNED BY:	B. JOHNSON
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US ARMY CORPS OF ENGINEERS
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SHEET LEGEND

DEFENSE SUPPLY CENTER RICHMOND (DSCR)
BUILDING 46 GENERATOR TIE-IN

DEMOLITION ONE-LINE DIAGRAM

SHEET ID
BLDG 46
E-601



